

Robust Crop Row Detection using Discrete Fourier Transform (DFT) for Vision-based In-Field Navigation

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Abstract

Accurate row detection is crucial for autonomous navigation systems in agricultural applications. Since the crops are typically planted in evenly spaced, parallel rows, crop rows mathematically correlate to a specific spectral signature within the frequency domain. In this study, we propose a robust frequency domain-based image processing pipeline for accurately locating crop rows within images. Using the two-dimensional discrete Fourier transform (DFT), we identify the spacing and direction of crop rows from significant frequency components, and their positions from the phase. To enhance precision, we apply frequency domain interpolation, a Hanning window, and discrete-time Fourier transform (DTFT) to accurately locate peak responses and corresponding phases in the spectrum. Extensive experiments were conducted to validate the accuracy and efficiency of our algorithm. The algorithm accurately detected crop rows in images and estimated lateral position and heading deviations relative to row centerlines, across different plant growth stages and illumination conditions. Additionally, by integrating a linear quadratic Gaussian (LQG) navigation controller with this vision-based crop row detection algorithm, we successfully enabled a robot to follow crop row centerlines with a mean absolute tracking error of 3.74 cm. These results highlight the potential of our frequency domain-based approach for enhancing precision agriculture through robust crop row detection and navigation.

Keywords: agricultural robot navigation; row detection; above-canopy imaging; discrete Fourier transform; LQG control

1. Introduction

1.1 General Background in Agricultural Field Navigation

Robotic techniques have been widely applied in various field operations in agriculture, including cultivation, spraying, harvesting, and weeding (Åstrand & Baerveldt, 2005). A robust navigation system is essential for robots to traverse between crop rows without damaging the crops.

Accurate estimation of robot poses relative to crop rows, including both position and heading, is fundamental for reliable in-field navigation. Currently, most agricultural navigation systems rely on global navigation satellite system/inertial navigation system (GNSS/INS) or machine vision systems for in field positioning and heading estimation (Wang et al., 2022).

GNSS/INS can provide centimeter-level positioning accuracy and precise heading estimations in open fields. However, GNSS/INS-based navigation requires a pre-planned operating trajectory

for path tracking. Additionally, GNSS signal quality may degrade under plant canopies or in greenhouses, resulting in insufficient positioning accuracy (Chen et al., 2020; Sivakumar et al., 2021). Moreover, the accuracy of INS is affected by temperature variations and integration drifts, impacting velocity and heading estimation (Shen et al., 2021). These limitations constrain the application of GNSS/INS-based navigation in certain scenarios.

Vision-based navigation is increasingly applied in in-field robotic applications due to its wide applicability, low cost, and high level of artificial intelligence. Visual guidance, which keeps a robot centered in constrained spaces like corridor-like environments, can be effectively applied to agricultural settings where crop rows act as parallel obstacles (Winterhalter et al., 2021). The fundamental technique in visual guidance is vision-based crop row detection. Top-view or front-view monocular color cameras are the most widely used vision sensors for crop row detection in visual guidance applications due to their reliability, affordability, and ability to capture semantic information (Sivakumar et al., 2021). However, achieving accurate crop row detection is challenging due to inconsistent and extreme illumination, weed infestation, and missing plants in crop fields.

1.2 Crop Row Detection Algorithms

The crop row detection algorithm must meet the following criteria for visual guidance applications: 1) accuracy with centimeter-level precision; 2) computational efficiency to maintain sufficient control rates; 3) capability to detect crop plants that may be smaller than weeds; and 4) robustness to function under high weed pressure and in the presence of missing plants within rows (Åstrand & Baerveldt, 2005). Various methods have been reported effective in identifying crop rows using a monocular camera. Based on the techniques used, these methods can be broadly categorized as “traditional methods” and “deep learning-based methods”.

1.2.1 Traditional methods

Traditional methods for crop row detection primarily locate crop plants in images and then fit lines corresponding to crop rows. Color-based segmentation algorithms are often applied, including color index-based segmentation, thresholding-based segmentation, and learning-based segmentation (Bai et al., 2023). However, these methods struggle with robustness due to uncontrolled lighting in the field. To improve robustness, some feature extraction algorithms were developed and applied, focusing on the morphological differences between crop plants and other objects like weeds. For instance, Harris corner features were used to determine the 3D position of crop rows in a binocular vision-based multi-crop row detection algorithm (Zhai et al., 2016). SUSAN corner features were extracted for corn seedling belt localization for a weed control robot (Lai et al., 2023). General feature extraction algorithms such as SIFT and FAST have also been effective in identifying crop rows in the field (Song et al., 2023; Y. Yang et al., 2023).

After extracting crop plants and features, line detection algorithms are often applied to fit line segments representing crop rows. The Hough transform is a widely used algorithm for detecting discontinuous lines in images (Bossu et al., 2006; Choi et al., 2015; Winterhalter et al., 2018), though it is computationally intensive. Variants like the median point Hough transform (Chen et al., 2020) and the probabilistic Hough transform (Lin et al., 2020) have been developed to improve efficiency and adaptability. Least square methods are less computationally intensive and have been used to fit straight lines as crop rows after semantic segmentation and feature extraction (Ospina & Noguchi, 2019; Liang et al., 2022). Other effective row detection algorithms include RANSAC and green pixels accumulation, etc. (García-Santillán et al., 2018; Li et al., 2020).

In general, traditional algorithms effectively detect crop rows by focusing on spatial information within the image, such as color, texture, and morphological features representing crop plants. However, the design and selection of handcrafted features require prior knowledge of the crop field. Line detection is also susceptible to noise caused by weed infestations and image defects due to extreme illumination, resulting in limited robustness.

1.2.2 Deep learning-based methods

With the development of artificial intelligence, deep learning techniques have been widely adopted in agricultural applications. Convolutional Neural Networks (CNNs), which can be trained to extract abstract features from images, are more robust than traditional algorithms against variations in crop plant shape, presence of weeds, and illumination inconsistencies in the field. The YOLO series are representative algorithms for object detection, where the bounding boxes of crop plants are extracted, and centroids are taken as feature points for crop plants. For instance, Yang et al. (2023) used the YOLOv5 network to extract centroids of maize and rice crop plants' bounding boxes, followed by a least square algorithm to find crop rows. Similarly, Liu et al. (2024) detected pineapple plants in fields using an enhanced YOLOv5 network for crop row detection. Diao et al. (2023) improved the YOLOv8 network to create the ASPPF-YOLOv8s network for locating the center of corn plants.

Semantic segmentation is another approach to accurately locating crop plants in images by labeling all pixels belonging to crop plants. For example, Yang et al. (2023) used the SegNet network to segment wheat canopy areas for detecting the centerline of wheat, while De Silva et al. (2024) employed U-Net to identify sugar beet crop regions and used the pixels to find crop row centerlines. Zhang et al. (2023) proposed the ResC-UNet network to improve maize crop row segmentation accuracy.

Beyond CNNs, Transformer networks have been leveraged to detect sugarcane rows and gaps, demonstrating better generalization to unseen data (dos Santos Ferreira et al., 2022). Some studies employed end-to-end CNN models to predict row and column anchors, simplifying the

crop row detection process without pixel-level segmentation or objective detection and improving adaptability (Quan et al., 2024).

In general, deep learning-based approaches have demonstrated significant superiority in crop plant detection applications. However, their performance is highly dependent on a high-quality annotated dataset. Building such a dataset requires substantial resources to collect diverse and high-quality images and accurately label them, which is both time-consuming and costly (Xiong et al., 2019). Beyond this, deploying deep learning models on field robots requires substantial hardware capabilities, which adds to the overall cost and complexity. These factors can hinder the widespread adoption of deep learning techniques for crop row detection in practical agricultural settings (S. Zhang et al., 2024).

1.2.3 Frequency domain analysis

Most of the aforementioned algorithms identify individual crop rows as separate lines within images rather than leveraging the global spatial pattern of the crop rows. This can result in false positive detections, particularly in situations where the plants are large (Winterhalter et al., 2018), the crop rows are infested with weeds, and there are missing plants (Romeo et al., 2012).

Given that agronomically planted crop rows are parallel and equally spaced with known spacing, this global spatial pattern imparts distinct and dominant properties in the 2D frequency domain. Weed infestations, missing plants, and image defects contribute to the noise at different frequencies than the crop rows. Frequency-based filtering algorithms have been developed to suppress the effects of weeds and missing plants for crop row detection (Hague & Tillett, 2001; Tenhunen et al., 2019). Specifically, the row patterns create a strong spectral signature where the frequency corresponds to row spacing, the location of such corresponding frequency in the 2D frequency domain represents direction, and its phase indicates the locations of the crop rows in the spatial domain. These spectral properties are not only true for straight crop rows but also valid for curved crop rows as they are largely linear within an image frame. Although there were studies using 2D Fourier transform and Gabor filter to locate parallel crop rows in aerial images, they were concerned about how to segment crop plants with pixel level accuracy, which was not designed to extract navigation lines for navigation applications (Vioix et al., 2002; Bossu et al., 2009).

Despite the aforementioned potential in the frequency-based approach in crop row detection, little research has been conducted on the feasibility of using frequency domain analysis, particularly the discrete Fourier transform (DFT), for detecting crop rows in agricultural field-based navigation applications. In this study, our overall goal was to explore the potential of applying DFT and frequency domain analysis to address the challenge of crop row detection for in-field navigation. In addition, with the development of such a frequency information driven row detection algorithm, as a proof-of-concept, it would be interesting to demonstrate how well the row detection results can be utilized for robot navigation control in the field. Hence, the

specific objectives of this study were to (1) develop the row detection algorithm in frequency domain; (2) assess the robustness of the developed row detection algorithm under different growth stages, field conditions, and outdoor lighting conditions; and (3) evaluate the performance of a navigation control system utilizing the proposed row detection algorithm.

The rest of the paper is organized as follows. In Section 2, we describe the hardware and data acquisition process, followed by an introduction to the proposed crop row detection algorithm. Section 3 presents the results of the quantitative analysis of position and heading deviation estimation, as well as the robot navigation experiments. This section also discusses the capabilities and limitations of the proposed method in depth. Finally, the study is concluded in Section 4.

2. Materials and Methods

2.1 Hardware and Data Acquisition

PhenoBot 3.0 (Gai et al., 2021; Bao et al., 2021), a robot developed by Iowa State University, was used as the platform for data acquisition and evaluation of the proposed method. Designed to automatically acquire phenotypic data of tall plants such as corn and sorghum at close range, PhenoBot 3.0 features a narrow body design (0.51 m wide and 2.1 m long) and a central articulated steering mechanism, enabling it to traverse between narrow crop rows (Figure 1). The robot is equipped with a telescoping, self-balanced sensor mast that carries cameras for capturing close-range and side-view stereoscopic images of plants at different heights, in order to extract morphological traits of corn and sorghum plants across different growth stages, including stalk size, ear height, leaf angle, and tassel/panicle morphology (Xiang et al., 2021).

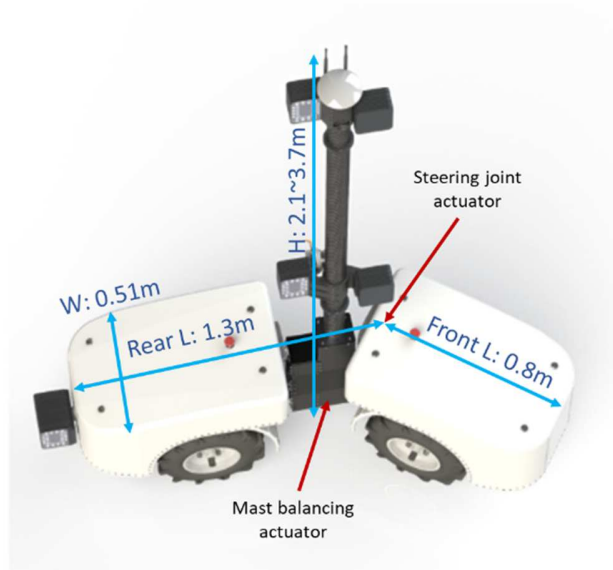


Figure 1. Phenobot 3.0 is articulated with a front and back section, and an actively-controlled vertical sensor mast for carrying sensors.

The images of crop rows used in this study were collected using a monocular RGB color camera (Phoenix 3.2MP, Lucid Vision Labs, Canada) placed on top of the sensor mast. With a 4.0 mm focal length lens, the camera has a horizontal view angle of 85.8° and a vertical view angle of 63.6° . The camera is angled downward at 37° and positioned 2.1 m above the ground when the mast is vertical and contracted. This setup allows the camera's field of view to cover crop rows from 1.1 m to 23 m in front of the robot, with the covered area width varying from 4 m (at 1.1 m ahead of the robot) to 40 m (at 23 m ahead of the robot). The raw images have a resolution of 2048×1536 pixels, with a spatial resolution of 2.0 mm/pixel at 1.1 m ahead of the robot and 21 mm/pixel at 23 m ahead of the robot (Figure 2).

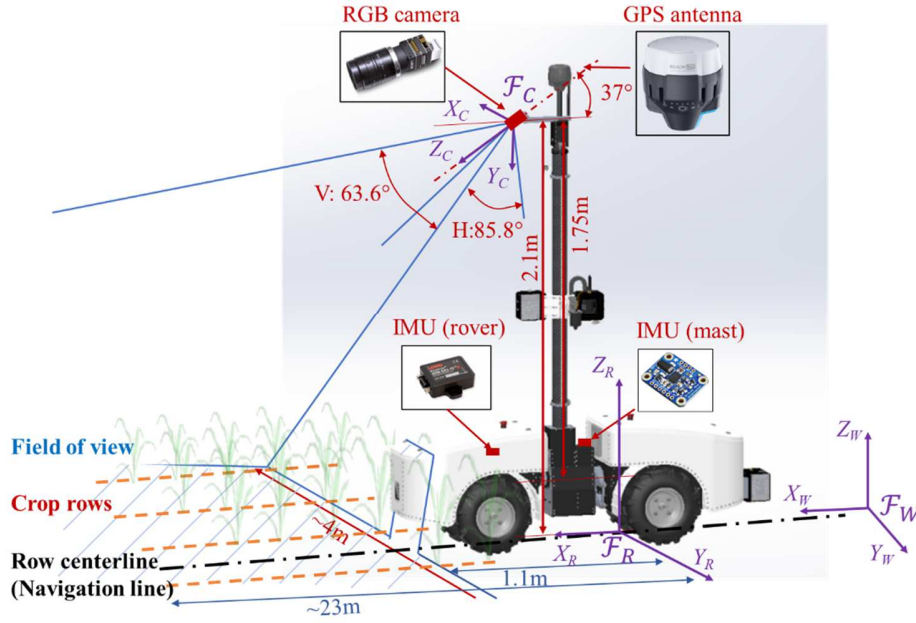


Figure 2. Illustration of the sensor placement, the field of view of the camera, and the definition of a series of coordinate frames used in the proposed algorithm and the navigation system, including the camera frame $\mathcal{F}_C$, the robot frame $\mathcal{F}_R$, and the world frame $\mathcal{F}_W$.

The ground truth of the robot pose in x , y and yaw , and the camera pose were calculated in real-time by fusing a series of sensors on the robot. An encoder measures the steering angle to calculate the robot posture change. An IMU (Adafruit BNO055, Adafruit, NY, USA) placed at the base of the sensor mast measures the incline angle of the mast. These two sensors track the pose of the camera frame $\mathcal{F}_C$ relative to the robot frame $\mathcal{F}_R$, denoted as T_R^C . The RTK-GPS antenna (Reach RS2, Emlid, Hungary) on the top of the sensor mast provides global positioning signal of the robot. The positioning accuracy is 7mm+1ppm in RTK mode. Motor encoders and a separate IMU (3DM-GX5-15, LORD Sensing MicroStrain, USA) on the rover were used for vehicle dead-reckoning and heading tracking. The bias instability of the accelerometer and the gyroscope of the IMU are 0.04 mg and $8^\circ/\text{hr}$, respectively. The signals from the RTK-GPS, IMU and motor encoders were fused by an Extended Kalman Filter (EKF) (Moore & Stouch, 2016) to

calculate the pose of the robot frame $\mathcal{F}_R$ relative to the world frame $\mathcal{F}_W$, denoted as T_W^R . The real-time estimation of the poses of the robot and the camera was implemented using the *TF2* and the *navigation* packages in ROS (Foote, 2013).

The images were collected in the fall of 2020 at an experimental cornfield within Iowa State University Agricultural Engineering and Agronomy Research Farm, located in Boone, Iowa, USA. The experimental field contained 12 crop rows, each about 50 m long, with a 76 cm (30") row spacing and about 15 cm (6") inter-plant spacing (Figure 3). The study considered varied working conditions, including crop growth stages, sunlight directions, and robot positions relative to the field edge. Data acquisitions were conducted when the corn plants were at the V3 and V7 growth stages. The sunlight illuminance ranged from 20,000 to 80,000 lux, and the camera exposure parameters were adjusted accordingly to ensure image quality. Images were taken at noon (~55° solar altitude angle) and evening (~20° solar altitude angle). Some images were taken when the robot faced the sunlight, causing glare, while others were taken when the robot moved along the sunlight direction, resulting in shadows of the robot in the images (Figure 4). The robot was manually driven at a speed of 0.8 m/s during data collection, and images were collected at 4 frames per second. The traveled paths during data acquisition are shown in Figure 3.

During image acquisition, the ground truth of the robot poses was calculated using the EKF-based positioning approach described above in real-time. For the ground truth of row centerlines, since the corn plants were sown in straight lines, we manually surveyed the position of the start and end points of each crop row in the field using an RTK-GPS antenna. The row centerlines were then determined by calculating the centers of every two adjacent rows. The calculated poses and the images were recorded with their timestamps using the *ros_bag* package in ROS. In total, the travel distance was about 1,000 m, and 6,755 images were recorded.

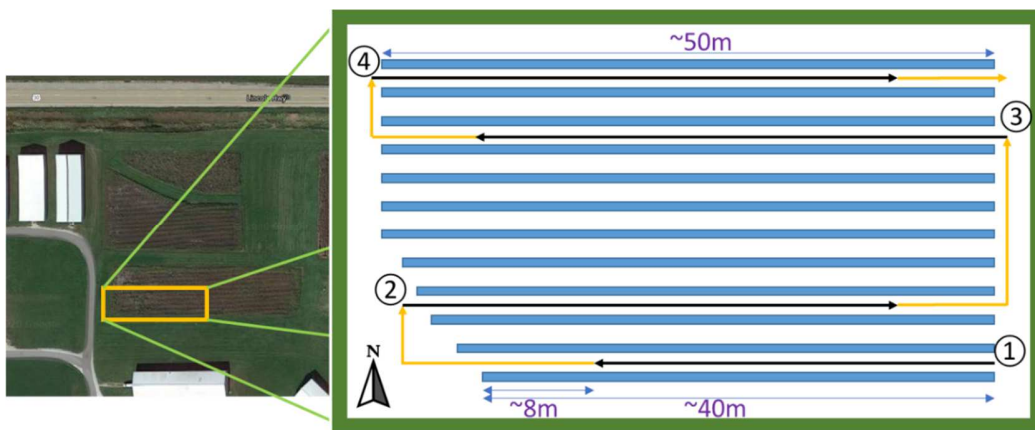


Figure 3. Robot paths for both data collection and navigation system evaluation in this study. The robot sequentially travelled through different crop rows, and the data collection and navigation system evaluation were performed when the robot was between crop rows and was more than 8 meters from the end of the row. The testing field was surrounded by grasslands and grown corn plants.



Figure 4. Images collected under different photographic conditions. The left image was collected at V3 growth stage, not at the edge of the field, glares exist in the image. The right image was collected at V7 growth stage, close to the edge of the field, shadows exist in the image.

2.2 Crop Row Detection and In-Row Pose Estimation

Vision-based row detection can be interpreted as the task of identifying parallel and periodic patterns in an image, despite the presence of perspective distortion and high-frequency noise. In this study, we proposed a DFT-based processing pipeline to efficiently extract the direction and position of crop rows from the frequency domain. The image processing pipeline consists of the following steps:

- 1) **Image Pre-processing:** Correct the lens distortion of the raw image and then transform the image into a simulated bird's-eye view using inverse perspective mapping to correct the perspective distortion.
- 2) **DFT-Based Crop Row Detection:** Transform the image into the two-dimensional frequency domain using fast Fourier transform (FFT) combined with a Hanning window function.
- 3) **Extraction of Crop Row Position and Orientation:** Detect the peak magnitude using Taylor expansion-based interpolation and calculate the corresponding phase angle with discrete time Fourier transform (DTFT).
- 4) **Estimation of Tracking Position and Heading Deviation:** Use the row detection results to estimate the robot's tracking position and heading deviation.

2.2.1 Image pre-processing

To make crop rows appear as parallel and periodic patterns in the image, optical distortion was corrected using distortion coefficients obtained from camera calibration (Z. Zhang, 2000). Perspective distortion was rectified by generating a bird's-eye view map through inverse

perspective mapping, projecting the image onto the ground plane (Figure 5), which is an inverse perspective mapping problem (Hartley & Zisserman, 2003). The transformation is represented by a 3x3 homography matrix H . The relationship between world coordinates on the plane and image coordinates can be expressed in a homogenous form:

$$s \begin{pmatrix} x_{\{I\}} \\ y_{\{I\}} \\ 1 \end{pmatrix} = H \begin{pmatrix} x_{\{M\}} \\ y_{\{M\}} \\ 1 \end{pmatrix} \quad (1)$$

Where $(x_{\{I\}}, y_{\{I\}})$ is a point in the image frame $\mathcal{F}_I$, $(x_{\{M\}}, y_{\{M\}}, 0)$ is the corresponding projected point in the local map frame $\mathcal{F}_M$ on the ground plane, and s is a scale factor. The local map frame $\mathcal{F}_M$ aligns with the robot frame $\mathcal{F}_R$, with the origin located at the robot's navigation reference position, and the $X_M - Y_M$ plane aligned with the ground plane.

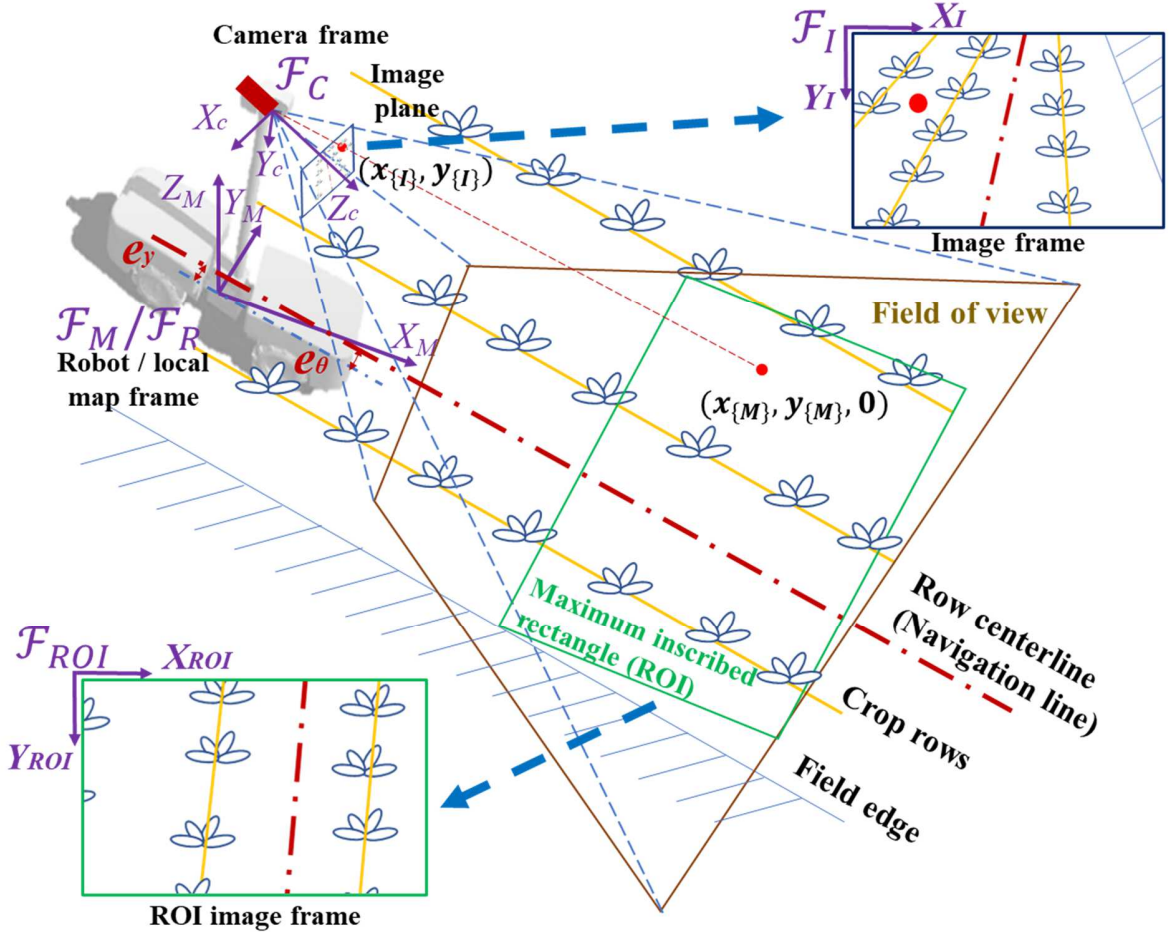


Figure 5. The illustration of the Homography transformation from the image frame $\mathcal{F}_I$ to a bird-eye's view image in ROI image frame $\mathcal{F}_{ROI}$. The relationship between a series of coordinate frames are illustrated, including the local map frame $\mathcal{F}_M$, the robot frame $\mathcal{F}_R$, the camera frame $\mathcal{F}_C$, the image

260 frame $\mathcal{F}_I$, and the ROI image frame $\mathcal{F}_{ROI}$. The definition of tracking deviation, including e_y for position
 261 deviation, and e_θ for heading deviation, are also illustrated.

262 The homography matrix H can be calculated using the intrinsic and extrinsic parameters of the
 263 camera. The intrinsic parameters are denoted by the intrinsic matrix $\mathbf{K}$, which defines the
 264 transformation between the image frame $\mathcal{F}_I$ and camera frame $\mathcal{F}_C$:

$$\mathbf{K} = \begin{bmatrix} f_x & 0 & c_x \\ 0 & f_y & c_y \\ 0 & 0 & 1 \end{bmatrix} \quad (2)$$

265 Where (f_x, f_y) are the focal lengths, and (c_x, c_y) are the coordinates of the principal point,
 266 obtained through camera calibration (Z. Zhang, 2000). The coordinates $(x_{\{I\}}, y_{\{I\}})$
 267 and $(x_{\{C\}}, y_{\{C\}}, z_{\{C\}})$ represent the same point in the image frame $\mathcal{F}_I$ and the camera frame $\mathcal{F}_C$,
 268 respectively.

$$s \begin{pmatrix} x_{\{I\}} \\ y_{\{I\}} \\ 1 \end{pmatrix} = \mathbf{K} \begin{pmatrix} x_{\{C\}} \\ y_{\{C\}} \\ z_{\{C\}} \end{pmatrix} \quad (3)$$

269 The extrinsic parameters of the camera, denoted as $\mathbf{T}_M^C$, represent the real-time pose of the
 270 camera frame $\mathcal{F}_C$ in the map frame $\mathcal{F}_M$, is calculated using sensors on the robot as described in
 271 Section 2.1. Therefore, the transformation matrix from the map frame to the camera frame, $\mathbf{T}_C^M$,
 272 is the inverse of $\mathbf{T}_M^C$:

$$\begin{pmatrix} x_{\{C\}} \\ y_{\{C\}} \\ z_{\{C\}} \\ 1 \end{pmatrix} = \mathbf{T}_C^M \begin{pmatrix} x_{\{M\}} \\ y_{\{M\}} \\ z_{\{M\}} \\ 1 \end{pmatrix} = (\mathbf{T}_M^C)^{-1} \begin{pmatrix} x_{\{M\}} \\ y_{\{M\}} \\ z_{\{M\}} \\ 1 \end{pmatrix} \quad (4)$$

273 Where $(x_{\{M\}}, y_{\{M\}}, z_{\{M\}})$ and $(x_{\{C\}}, y_{\{C\}}, z_{\{C\}})$ are the coordinates of the same point in the local
 274 map frame $\mathcal{F}_m$ and the camera frame $\mathcal{F}_C$, respectively.

275 Based on the principle of projective projection, for a point on the ground plane $(x_{\{M\}}, y_{\{M\}}, 0)$ in
 276 the local map frame $\mathcal{F}_M$ and the corresponding point $(x_{\{I\}}, y_{\{I\}})$ on the image plane, the
 277 relationship is:

$$s \begin{pmatrix} x_{\{I\}} \\ y_{\{I\}} \\ 1 \end{pmatrix} = \mathbf{K} \cdot [\mathbf{I} | \mathbf{0}] \cdot \mathbf{T}_C^M \begin{pmatrix} x_{\{M\}} \\ y_{\{M\}} \\ 0 \\ 1 \end{pmatrix} \quad (5)$$

278 Where $\mathbf{I}$ is a 3x3 identity matrix. When representing $\mathbf{T}_C^M$ as:

$$\mathbf{T}_C^M = \begin{bmatrix} \mathbf{R}_{3 \times 3} & \mathbf{T}_{3 \times 1} \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} r_{11} & r_{12} & r_{13} & t_x \\ r_{21} & r_{22} & r_{23} & t_y \\ r_{31} & r_{32} & r_{33} & t_z \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (6)$$

Where r_{ij} is the element of the rotation matrix $\mathbf{R}$, and t_x, t_y, t_z are the translations along the corresponding axis. The homography matrix $\mathbf{H}$ in equation (1) is therefore:

$$\mathbf{H} = \mathbf{K} \cdot \begin{bmatrix} r_{11} & r_{12} & t_x \\ r_{21} & r_{22} & t_y \\ r_{13} & r_{23} & t_z \end{bmatrix} \quad (7)$$

Thus, the inverse perspective mapping from the image frame $\mathcal{F}_I$ to the local map frame $\mathcal{F}_M$ can be formulated as:

$$\begin{pmatrix} x_{\{M\}} \\ y_{\{M\}} \\ 1 \end{pmatrix} = \frac{1}{s} \mathbf{H}^{-1} \begin{pmatrix} x_{\{I\}} \\ y_{\{I\}} \\ 1 \end{pmatrix} \quad (8)$$

where s is a scale factor to normalize the homogeneous coordinates after applying the transformation.

After generating the bird's eye view image, the maximum axis-aligned inscribed rectangle (MAAIR) of the projected area was extracted as a new region of interest (ROI) image (Figure 5). The MAAIR ensures all pixels in the extracted area are valid (i.e. has the corresponding pixels in the raw image) and contain the most information for crop row detection. The area can be further cropped based on the surveyed field boundary. To ensure accurate row detection, the maximum looking ahead distance was set to 16 m for heading estimation and 8 m for position deviation estimation. The spatial resolution of the ROI image was set to 0.01 m/pixel. The transformation between the local map frame $\mathcal{F}_M$ and ROI image frame $\mathcal{F}_{ROI}$, which is simply a combination of scaling and translation on ground plane, was recorded. The extracted ROI image was converted into grayscale using the Excess Green Index (ExG) (W. Yang et al., 2015) to highlight the vegetation pixels.

2.2.2 Hanning window for Fourier transform

According to the intrinsic of Fourier transformation, the crop row pattern in the ROI image corresponds to a peak in the low frequency part of the spectrum. The period (i.e. the inverse of the frequency) of the peak represents the spacing, the angular location of that peak represents the direction of the crop rows, and the corresponding phase angle represents the position of the crop rows. However, directly applying the two-dimensional FFT algorithm to the image and finding the bin with maximum magnitude in the spectrum is inaccurate in locating the crop rows. This inaccuracy is due to insufficient resolution for low-frequency components and the spectrum

leakage problem of the 2D FFT (Weichert et al., 2023). In this study, a Hanning window was applied to the image during DFT to reduce spectrum leakage.

The principle is as follows: the image boundary inherently acts as a rectangular window, which contributes to spectrum leakage. Applying a Hanning window is a widely used approach to reduce spectrum leakage and increase the accuracy of peak detection in signal processing (Ming & Kang, 1996). For 1D signals, such as time-series data, the Hanning window function $w(n)$ is defined as:

$$w(n) = 0.5 - 0.5 \cos\left(\frac{2\pi n}{M-1}\right), 0 \leq n \leq M-1 \quad (9)$$

where M is the sample size and n is the sampled position (i.e., the timestamps of a time-series data). Compared with a rectangular window, the frequency domain of Hanning has lower side lobes, which are responsible for spectral leakage (Figure 6). The reduction in side lobe levels leads to a cleaner frequency spectrum with less interference from adjacent frequencies. Therefore, the Hanning window increases the robustness of the DFT result compared with a rectangular window, thus reducing spectral leakage.

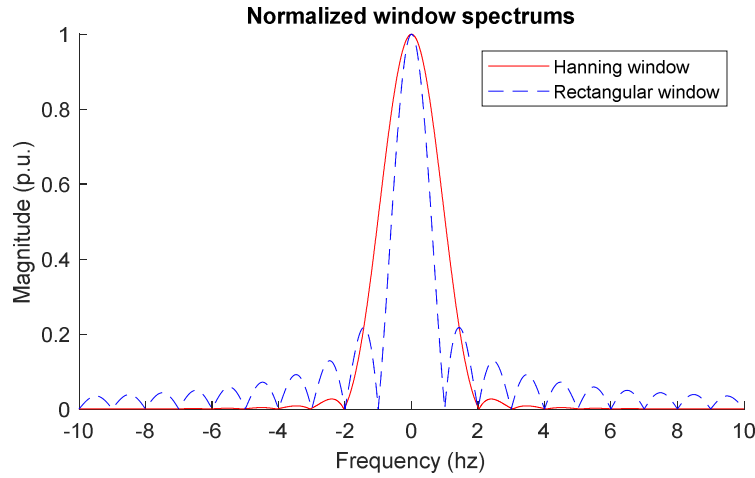


Figure 6. The frequency spectrum of a rectangular window and a Hanning window. The Hanning window has a wider main lobe, and lower side lobes.

In this study, for each ROI image acquired by the previous steps, the 2D Hanning window function $w(x, y)$ was generated as the cross product of two 1D Hanning window functions:

$$w(x, y) = \frac{1}{4} \left[1 - \cos\left(\frac{2\pi x}{M-1}\right) \right] \left[1 - \cos\left(\frac{2\pi y}{N-1}\right) \right], \begin{cases} 0 \leq x \leq M-1 \\ 0 \leq y \leq N-1 \end{cases} \quad (10)$$

where (M, N) is the image size and (x, y) is the sampled pixel position (i.e. image pixel coordinate along each axis). After applying the 2D Hanning window to the image, the FFT algorithm was used to compute the magnitude spectrum $f(x, y)$ and the phase spectrum $p(x, y)$.

2.2.3 Crop row pattern detection and localization

To locate crop rows with sufficient accuracy for robot navigation, sub-pixel resolution is necessary for identifying the magnitude peak in the magnitude spectrum $f(x, y)$. The steps are as follows:

1) Bandpass filtering and coarse peak detection: a bandpass filter is firstly applied to the magnitude spectrum $f(x, y)$ to preserve only the components with periods closed to the crop row distancing. And the magnitude peak is coarsely located as the maximum magnitude value in the spectrum $f(x, y)$, with its position recorded as (x_0, y_0) .

2) Refining peak detection using Taylor expansion: in the magnitude spectrum $f(x, y)$, a Taylor expansion-based interpolation is applied at (x_0, y_0) to refine the peak detection result as $(u, v) = (x_0 + \Delta x, y_0 + \Delta y)$, where $(\Delta x, \Delta y)$ is the position bias. Since the gradient of a peak is zero by definition, the bias $(\Delta x, \Delta y)$ can be computed by solving the following equation:

$$\begin{pmatrix} f'_x(x_0 + \Delta x, y_0 + \Delta y) \\ f'_y(x_0 + \Delta x, y_0 + \Delta y) \end{pmatrix} = \begin{pmatrix} f'_x(x_0, y_0) \\ f'_y(x_0, y_0) \end{pmatrix} + \begin{pmatrix} f''_{xx}(x_0, y_0) & f''_{xy}(x_0, y_0) \\ f''_{xy}(x_0, y_0) & f''_{yy}(x_0, y_0) \end{pmatrix} \begin{pmatrix} \Delta x \\ \Delta y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \quad (11)$$

The 1st and 2nd order derivatives of $f(x, y)$ were computed numerically. Therefore, the bias $(\Delta x, \Delta y)$ is:

$$\begin{pmatrix} \Delta x \\ \Delta y \end{pmatrix} = - \begin{pmatrix} f''_{xx}(x_0, y_0) & f''_{xy}(x_0, y_0) \\ f''_{xy}(x_0, y_0) & f''_{yy}(x_0, y_0) \end{pmatrix}^{-1} \begin{pmatrix} f'_x(x_0, y_0) \\ f'_y(x_0, y_0) \end{pmatrix} \quad (12)$$

3) Phase angle calculation using DTFT: the corresponding phase φ of the magnitude peak is determined by applying the discrete time Fourier transform (DTFT) on the windowed ROI image obtained in Section 2.2.2. The frequency content of the ROI image signal at the frequency of (u, v) was calculated using DTFT:

$$F(u, v) = \sum_{m=0}^M \sum_{n=0}^N f(m, n) e^{-j2\pi(um/M + vn/N)} \quad (13)$$

where (m, n) are the pixel coordinates in the ROI image, and (M, N) is the image size. The accurate estimation of the phase angle φ is:

$$\varphi = \text{phase}(F(u, v)) \quad (14)$$

Since DTFT is more computationally intensive compared to FFT, it is only applied once for accurate phase estimation after locating the peak frequency.

4) Determining crop row positions: the positions of the crop rows in the image are the wavefronts of the sinusoidal waves corresponding to the peak frequency (u, v) . The points on

349 the wavefront lines have their phase values equal to 0. Therefore, the periods T_x , T_y of the crop
 350 rows along x and y directions are:

$$T_x = \frac{M}{u} \quad (15)$$

$$T_y = \frac{N}{v} \quad (16)$$

351 where (M, N) is the image size, and units are in pixels. Therefore, the normalized line direction
 352 vector $\mathbf{l}$ of the crop rows in the ROI image is:

$$\mathbf{l} = (l_x, l_y) = (-T_x, T_y) / \|(-T_x, T_y)\| \quad (17)$$

353 A set of intersections $\mathbf{P} = \{\mathbf{p}_1, \mathbf{p}_2, \dots\}$ of the parallel lines along the x-axis in the image
 354 coordinate were calculated as:

$$\mathbf{p}_k = [T_x * (2k\pi - \varphi), 0], k \in \mathbf{Z} \quad (18)$$

355 Therefore, with the determined set of intersections $\mathbf{P}$ and the line direction $\mathbf{l}$, the position and
 356 direction of the crop rows are established in the ROI image frame $\mathcal{F}_{ROI}$.

357 2.2.4 Position and heading deviation estimation

358 Lastly, the tracking deviation and heading deviation were determined. Specifically, the detected
 359 parallel crop rows in the ROI image frame $\mathcal{F}_{ROI}$ were mapped back to the local map frame $\mathcal{F}_M$.
 360 For instance, for a crop row which has a direction vector of $\mathbf{l}$ passing through $\mathbf{p}_k$, the mapped
 361 line in the local map frame $\mathcal{F}_M$ becomes:

$$\mathbf{l}_{\{M\}} = (l_{x\{M\}}, l_{y\{M\}}, 0)^T = \text{normalize}(\mathbf{T}_M^{ROI}(l_x, l_y, 0)^T) \quad (19)$$

$$\mathbf{p}_{k\{M\}} = (x_{k\{M\}}, y_{k\{M\}}, 1)^T = \mathbf{T}_M^{ROI}(T_x * (2k\pi - \varphi), 0, 1)^T \quad (20)$$

362 Then the navigation line to track was determined as the centerline of two adjacent crop rows that
 363 have the shortest perpendicular distance to the robot position, which is the origin in the local map
 364 frame $\mathcal{F}_M$ (Figure 5). The signed perpendicular distance d_k from the robot position to the crop
 365 rows defined by $\mathbf{l}_{\{M\}}$ and $\mathbf{p}_{k\{M\}}$ was calculated by a dot product:

$$d_k = (x_{k\{M\}}, y_{k\{M\}}) \cdot (l_{x\{M\}}, l_{y\{M\}}) \quad (21)$$

366 The tracking deviation e_y was calculated as the perpendicular distance from robot position to the
 367 centerline of two lines with the shortest perpendicular distance, d_k and d_{k+1} , to the robot, as
 368 illustrated in Figure 5:

$$e_y = (d_k + d_{k+1})/2 \quad (22)$$

369 This metric quantifies the lateral displacement of the vehicle from the desired path. The heading
 370 deviation e_θ is defined as the angular difference between the vehicle's heading direction and the
 371 direction of the navigation line. It quantifies the misalignment of the vehicle's heading with
 372 respect to the row direction.

$$e_\theta = -\text{atan}(l_{x\{M\}}/l_{y\{M\}}) \quad (23)$$

373 The row detection algorithm was implemented in Python using *OpenCV* and *Eigen* library.

374 **3. Results and Discussions**

375 The experiments were designed to evaluate the performance of our proposed navigation control
 376 system for in-field navigation, specifically focusing on (1) the performance of the proposed
 377 DFT-based algorithm in locating crop rows and (2) the performance of the row detection
 378 algorithm within a row-guidance system.

379 **3.1 Row Detection Algorithm Evaluation**

380 This experiment evaluated the proposed method's ability to find crop rows in images and
 381 estimate tracking deviations. The robustness of the algorithm was also assessed against various
 382 crop growth stages, sunlight directions, and robot positions in the field.

383 **3.1.1 Evaluation matrices**

384 The Pearson correlation coefficients (r) and root mean square errors (RMSE) of both calculated
 385 position and heading deviations (e_y , e_θ) were used as evaluation matrices for the proposed row
 386 detection algorithm. The definition of r and RMSE are as follows:

$$r = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^n (x_i - \bar{x}) \sum_{i=1}^n (y_i - \bar{y})}} \quad (24)$$

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (x_i - y_i)^2} \quad (25)$$

387 Where x_i and y_i are the predicted values and the corresponding ground truth values, respectively.
 388 $\bar{x}$ and $\bar{y}$ are the means of x_i and y_i , and n is the number of data samples.

389 **3.1.2 Qualitative analysis**

390 Among the 6755 collected images, 6264 images were successfully processed, and the crop rows
 391 were detected. The average processing time for a frame was 252 ms with a standard deviation of

33 ms, using a computer with an i5-6500TE processor. Most failures occurred when the robot traversed hard bumps in the field, causing rapid camera pose changes that the selected IMU model on the mast could not track, leading to incorrect camera pose estimations and rectification results.

The step-by-step results of the proposed DFT-based row detection algorithm are shown in Figure 7 to visualize its effectiveness. The algorithm successfully revealed the crop rows within the images at different growth stages and proved robust to missing crop rows, glare effects, and shadows, which are significant challenges for crop row detection in the field.

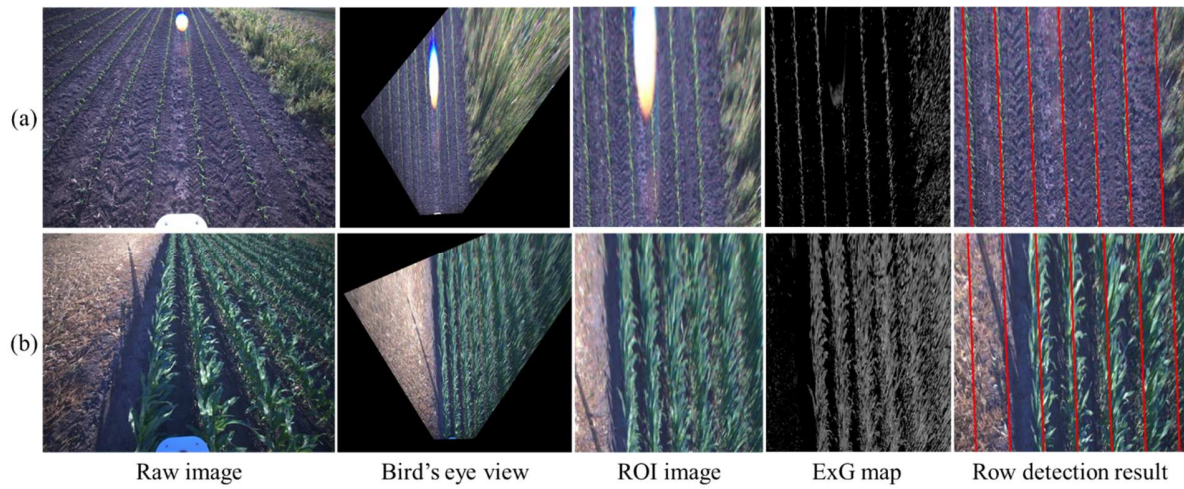


Figure 7. Step results of the proposed DFT-based row detection algorithm: (a) image taken at V3 growth stage, not at the edge of the field, with glares. (b) image taken at V7 growth stage, at the edge of the field, with shadows.

3.1.3 Performance analysis for different seniors

The proposed algorithm's performance in tracking deviation estimation was evaluated under different scenarios, defined according to the growth stages of the corn (GS), time of day (TD), and whether the robot was facing the sunlight (FS) or near the edge of the field (NE). As shown in Table 1, the average RMSE for lateral position and heading deviation estimation across all tested scenarios were 6.43 cm and 1.48°, respectively, with corresponding r values of 0.52 and 0.69.

Applying the algorithm to image sequences from entire passes during data collection showed that the tracking deviation estimation closely followed the ground truth, indicating the algorithm's suitability for in-field vision-based navigation (Figure 8).

Table 1. The r values and RMSEs for the estimation of position deviation (Pos Dev) and angular heading deviation (Ang Dev) using the proposed row detection algorithm in different scenarios. Scenarios were

417 defined according to the growth stages of the corn (GS), time of the day (TD), and whether they are:
 418 facing the sunlight (FS), and near the edge of the field (NE).

Scenarios				Pos Dev		Ang Dev	
GS	TD	FS	NE	r value	RMSE/cm	r value	RMSE/deg
V3	Noon	Y	Y	0.54	5.50	0.67	1.24
V3	Noon	N	N	0.68	6.28	0.85	1.40
V3	Noon	Y	N	0.48	4.50	0.48	1.30
V3	Noon	N	Y	0.45	4.26	0.59	1.31
V3	Evening	Y	Y	0.67	7.53	0.86	2.22
V3	Evening	N	N	0.40	7.43	0.67	1.68
V3	Evening	Y	N	0.44	5.97	0.66	1.04
V3	Evening	N	Y	0.33	7.09	0.64	1.07
V7	Noon	Y	Y	0.45	7.29	0.50	1.78
V7	Noon	N	N	0.67	5.25	0.79	1.59
V7	Noon	Y	N	0.46	7.00	0.49	1.86
V7	Noon	N	Y	0.46	5.96	0.76	1.65
V7	Evening	Y	Y	0.59	7.46	0.84	1.34
V7	Evening	N	N	0.44	7.64	0.75	1.45
V7	Evening	Y	N	0.61	6.48	0.72	1.50
V7	Evening	N	Y	0.60	7.22	0.84	1.32
Average				0.52	6.43	0.69	1.48

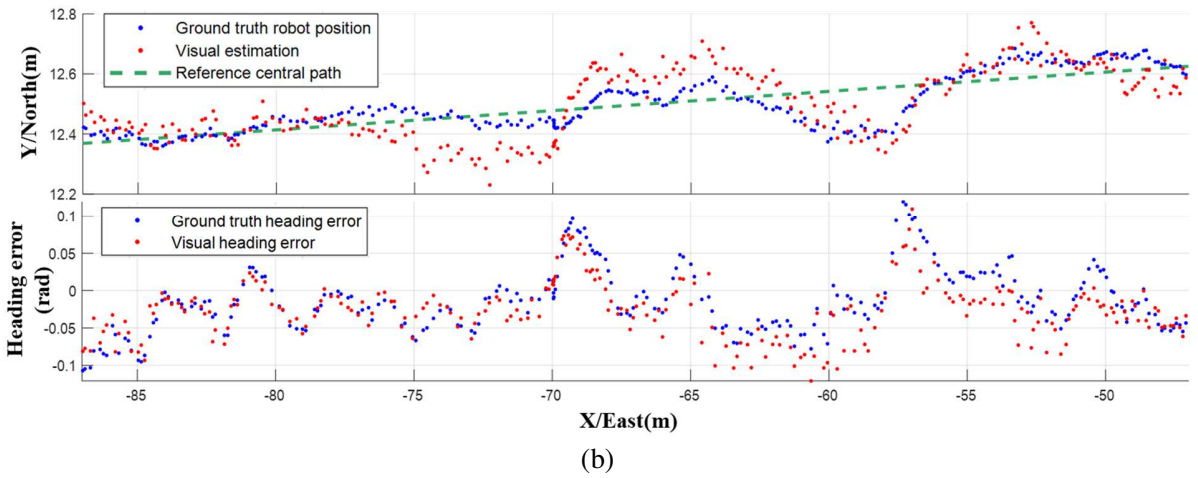
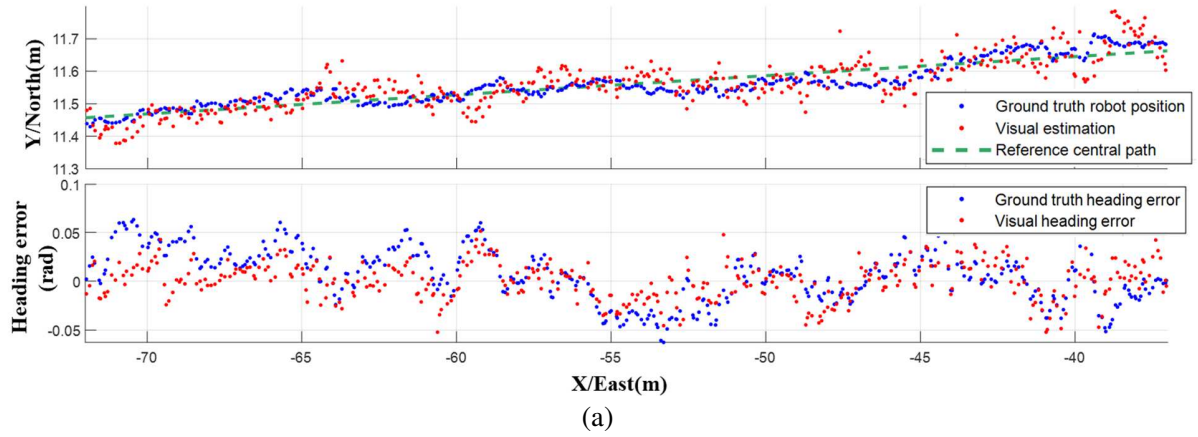


Figure 8. Tracking deviation estimation of example passes, including (top) the estimated robot trajectory calculated from position deviation, and (bottom) the measured heading deviation compared with ground truth from GPS-IMU localization. (a) V3 growth stage, evening time, facing the sun, not near the edge of the field; (b) V7 growth stage, evening time, at the edge of the field, with shadows.

As shown in Table 1, performance varied across different scenarios. To investigate factors affecting the accuracy of position and heading deviation estimation, a series of two-sample t-tests (Snedecor & Cochran, 1991) were conducted. The null hypothesis was that the means are equal, indicating that the accuracy of position or heading deviation estimation is not affected by the tested factor. As shown in Table 2, only one test had a significant p-value to reject the null hypothesis, indicating that the performance of lateral deviation estimation is affected by the time of day. Specifically, operating in the evening with an inclined sunlight angle reduced the accuracy of lateral position estimation. However, the difference in accuracy between different growth stages was not significant. Despite less obvious crop row patterns at later growth stages, heading and position deviations were correctly estimated due to accurate row direction estimation from the global pattern in the frequency domain and the higher accuracy in the center of images due to the Hanning window function.

Table 2. Equal mean tests and results of the RMSE's of the lateral position and heading deviation estimation by different factors.

No. of test	Response	Factor	p-value
1	Pos Dev RMSE	Growth stage (V3/V7)	0.132
2	Pos Dev RMSE	Time of the day (Noon/Evening)	0.011*
3	Pos Dev RMSE	Facing the sunlight (Y/N)	0.897
4	Pos Dev RMSE	Near the edge (Y/N)	0.702
5	Ang Dev RMSE	Growth stage (V3/V7)	0.345
6	Ang Dev RMSE	Time of the day (Noon/Evening)	0.681
7	Ang Dev RMSE	Facing the sunlight (Y/N)	0.520
8	Ang Dev RMSE	Near the edge (Y/N)	0.945

In general, the algorithm accurately detected crop rows and estimated the robot's tracking deviation. The proposed method is robust to varying plant growth stages, operation positions in the field, and illumination variations, including glares and shadows in the images. However, correct camera calibration is essential. The algorithm's performance decreases when the camera pose is inaccurately calculated, resulting in non-parallel and unevenly spaced crop rows in the simulated bird's-eye view.

3.2 Navigation System Evaluation

The second experiment evaluated the effectiveness of the proposed row detection algorithm within a navigation system for crop row tracking. A proof-of-concept navigation system was developed, utilizing an LQG controller along with visually derived tracking deviations for robot steering control.

3.2.1 Navigation control system setup

Since the position and heading deviation measurements using the DFT-based method are not perfect, the steering control system must accommodate input signals with a certain level of uncertainty. In this application, an LQG controller was applied to steer the robot, correcting lateral position and heading deviations for in-row navigation. The LQG controller combines a

458 Linear Quadratic Regulator (LQR) with a Kalman filter, which considers noise in the linear
 459 system and measurement process.

460 For this study, the general path tracking error model for car-like vehicles was used as the robot's
 461 kinematic model (Corke, 2011). The discrete system model with process and measurement noise
 462 is defined as:

$$\begin{aligned} \mathbf{x}_{n+1} &= \mathbf{A}\mathbf{x}_n + \mathbf{B}\mathbf{u}_n + \mathbf{w} \\ \mathbf{y}_n &= \mathbf{C}\mathbf{x}_n + \mathbf{v} \end{aligned} \quad (26)$$

463 in which $\mathbf{w}$ is the process noise vector with covariance matrix $\mathbf{Q}$, and $\mathbf{v}$ is the measurement noise
 464 vector with covariance matrix $\mathbf{R}$. The state vector $\mathbf{x}$ of the system includes the lateral position
 465 error e_y , the heading error e_θ and their rates. The input $\mathbf{u}$ is the turning curvature κ of the robot,
 466 controlled by actuating the steering joint. The measurement $\mathbf{y}$ are the lateral position error $\widehat{e}_y$,
 467 and the heading error $\widehat{e}_\theta$ acquired by the proposed row detection algorithm.

$$\mathbf{x} = (e_y, e_y', e_\theta, e_\theta')' \quad (27)$$

$$\mathbf{u} = (\kappa) \quad (28)$$

$$\mathbf{y} = (\widehat{e}_y, \widehat{e}_\theta)' \quad (29)$$

468 Based on the kinematic model of a car-like vehicle, the system model (Equation 26) becomes:

$$\begin{pmatrix} e_y \\ e_y' \\ e_\theta \\ e_\theta' \end{pmatrix}_{n+1} = \begin{pmatrix} 1 & \Delta t & 0 & 0 \\ 0 & 0 & s & 0 \\ 0 & 0 & 1 & \Delta t \\ 0 & 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} e_y \\ e_y' \\ e_\theta \\ e_\theta' \end{pmatrix}_n + \begin{pmatrix} 0 \\ 0 \\ 0 \\ v \end{pmatrix} \kappa + \mathbf{w} \quad (30)$$

469 in which s is the separately controlled forward speed, and Δt is the control cycle time. The
 470 covariance matrix $\mathbf{Q}$ of process noise $\mathbf{w}$ was designed according to the confidence level of the
 471 robot kinematic model. The covariance matrix $\mathbf{R}$ of measurement noise $\mathbf{v}$ was designed according
 472 to the RMSE values of the position and heading deviation estimation. With the designed LQG
 473 controller, a proof-of-concept navigation system was implemented (Figure 9).

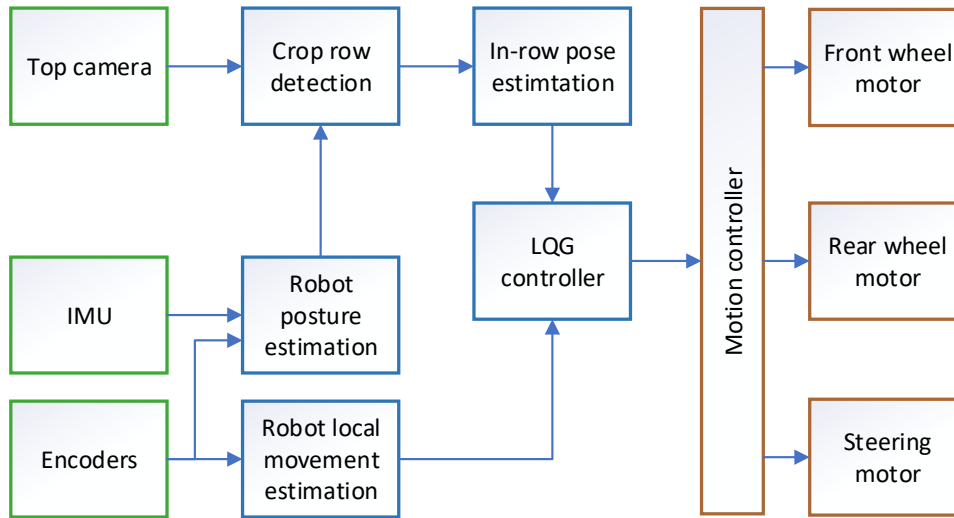


Figure 9. The architecture of the vision-based navigation control system for tracking crop rows

3.2.2 Field tests results

The field tests were conducted in the same field and the same passes used during image acquisition (Figure 3). Using PhenoBot 3.0 as a testing platform, the experiment was carried out with corn plants at the V3 growth stage during the evening. The actual robot trajectory was measured using GPS-based localization, and the position and heading deviations were analyzed. The forward speed was controlled at 0.8 m/s. The sampling rate of the camera for the row detection algorithm was 3 frames per second, and the update frequency of the LQG controller was set to 8 Hz.

Our experimental results showed that the trajectory along which the robot navigated closely matched the reference central path (Figure 10) with a reasonable distribution of the tracking error (Figure 11). The mean absolute tracking error was 3.74 cm in average, with the position error being less than 8.5 cm in 90% of the cases (Table 3). Among the four passes in the trajectory, lower tracking errors were observed when operating in the central region of the field compared to near the edges, with average RMSEs of 3.22 cm and 5.40 cm, respectively.

As a proof-of-concept, the results have demonstrated that when integrated with an LQG controller, the proposed row detection algorithm effectively guided the robot to maintain its position centered between crop rows, validating its applicability for field robot operations. The navigation control system exhibited robust performance under varying illumination conditions, including glares and shadows. However, the system's performance degraded when operating near field edges, primarily due to two factors: uneven terrain that increased the process noise level for the LQG controller, and increased errors in robot pose estimation.

For future implementation of the navigation control system in field conditions, we propose the following improvements: first, adopting a more robust strategy for robots pose estimation;

Second, implementation of an advanced controller capable of adapting to time-variant process noise when operating on uneven terrain; Third, development of a row-end-detection method to address the dilution of dominant frequency components from crop rows by non-crop areas at row ends. These limitations will be addressed in our future study.

Compared to navigation control systems using alternative row detection algorithms, our proposed system demonstrates superior robustness across varying plant growth stages, operation positions in the field, and illumination variations, including glares and shadows in the images. Furthermore, the major computational costs of our algorithm are concentrated in 2D DFT and DTFT calculations, which can be efficiently accelerated using commercial embedded digital signal processing (DSP) units. This enables deployment on embedded computational systems, offering lower costs and higher stability for field-based platforms such as agricultural robots and UAVs.

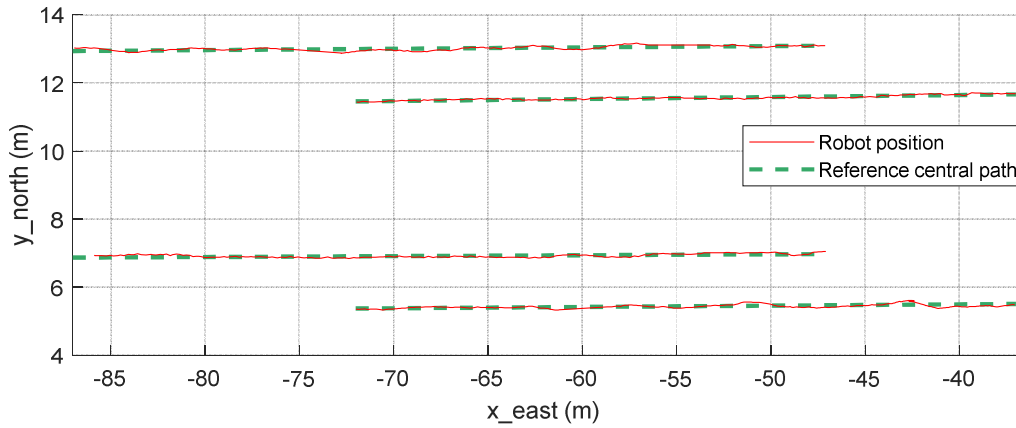


Figure 10. Trajectories of the robot using the proposed navigation system

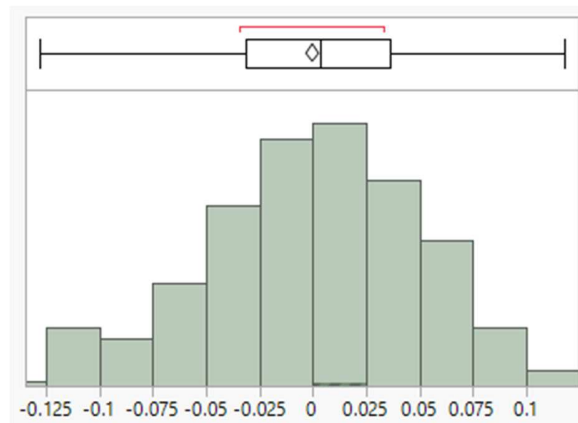


Figure 11. The distribution of the tracking error (lateral position), unit: m.

Table 3. The RMSEs and 90% percentile for the tracking error (lateral position relative to the reference central path) in each pass during the field tests. Scenarios were defined according to whether they are: facing the sunlight (FS), and near the edge of the field (NE).

No. of pass	Scenarios		Tracking error (cm)	
	FS	NE	RMSE	90% percentile
1	Y	Y	5.68	10.33
2	N	N	3.53	6.70
3	Y	N	2.65	4.18
4	N	Y	4.99	8.77
Average			3.74	8.50

In summary, the results of this study demonstrated that the proposed DFT-based row detection algorithm effectively detected crop rows, contributing to agricultural robot navigation in the field. Compared to other vision-based row detection algorithms, the benefits and limitations of the proposed DFT-based row detection algorithm are as follows:

Benefits:

- 1) The algorithm utilizes the parallel property and row spacing of the crop rows as prior knowledge to accurately locate the crop rows in the images.
- 2) By focusing on global frequency features in images, the approach is robust to various local noise, such as weeds, glares, and shadows.
- 3) The computational load for processing each image is low and relatively consistent, involving only a few configuration parameters.

Limitations and potential solutions:

- 1) The algorithm relies on accurate camera pose estimation for field-based navigation. Therefore, precise robot-camera calibration, and real-time sensing of robot pose and posture are required when applying this algorithm in field applications.
- 2) The algorithm has an RMSE of about 6 cm for lateral position deviation estimation. Therefore, the algorithm necessitates a controller compatible with noisy input signals, such as the LQG controller used in this study.

4. Conclusions

In this paper, we proposed a frequency domain-based crop row detection algorithm for agricultural robot navigation. Assuming sectionally straight, parallel, and equally-spaced crop rows, we developed a novel image processing pipeline incorporating Taylor expansion-based interpolation, Hanning windowing, and DTFT to enhance the accuracy of crop row detection in the frequency domain. Experiments conducted with corn plants at various growth stages, under different illumination conditions, and at different positions in the field demonstrated the algorithm's effectiveness. The results showed that the proposed crop row detection algorithm accurately estimated lateral position and heading deviations relative to the row center, with average RMSEs of 6.43 cm and 1.48°, respectively. Using an LQG controller, the robot maintained a mean absolute tracking error of 3.74 cm, with tracking errors less than 8.5 cm in 90% of cases.

Our study demonstrated that the DFT and frequency domain method is a viable approach for crop row detection in vision-based navigation within crop fields. By focusing on the parallel and equally spaced nature of crop rows, the algorithm proved robust against local noise such as weeds, glares, and shadows. However, the algorithm's performance depends on accurate robot-camera calibration and real-time sensing of robot pose and posture.

In summary, the presented algorithm and field-testing results confirmed the feasibility of using frequency domain analysis for robotic navigation between crop rows, while also highlighting limitations that warrant further investigation. The framework of the proposed algorithm can be extended to various agricultural applications involving above-canopy cameras, including mapping with unmanned aerial vehicles (UAVs).

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